

Regression Challenge: Final Analysis

Yuetao Li | Spring 2026 Kaggle Challenge Report

Methodology and Generalization Results

tl;dr: Main Idea and Results

Key Result

0.07075

Final Private R^2 Score

(Beating Baseline 0.04599)

The project successfully bypassed "noise traps" by isolating high-signal features and applying robust preprocessing. While the public score was low (0.00556), the model Generalized exceptionally well to the unseen private dataset.

- ✓ **Top Features:** x6 and x9 (Signal Discovery)
- ✓ **Key Strategy:** Outlier shielding + Polynomial interaction

EDA: Initial Feature Analysis

Signal vs. Noise

Initial analysis focused on the 15 provided features (x0 to x14). Correlation mapping revealed:

- **Feature x9:** Strongest direct correlation with the target.
- **Feature x6:** Weak individual correlation, but later identified as a critical suppressor variable.
- **Noise Columns:** Most features showed near-zero correlation, acting as pure noise traps for linear models.

EDA: Detecting the Noise Traps

High Dimensionality

The provided 15 features to encourage over-engineering.

Most models failed by trying to find patterns in garbage columns.

Trap Correlations

Certain features showed "fake" linear patterns in the training set that vanished in the Kaggle public leaderboard test set.

Linear Failure

Standard Linear Regression on all 15 features yielded a negative public R^2 , confirming the "Noise Trap" hypothesis.

EDA: Outlier Distribution in Target

Target Characteristics

Metric	Value
Min Target (y)	-41,008.02
Max Target (y)	+69,628.20
Std Dev	High (Noisy)

The massive range of the target variable indicated that Mean Squared Error (MSE) would be highly sensitive to extreme outliers.

Experiment 1: Full-Feature Failure

Standard Ridge + StandardScaler

Outcome: Public Score -0.07800

Motivation: Test if a baseline linear approach on the full feature set could survive. The result was a total failure. **Key Learning:** StandardScaler was insufficient for the extreme test-set outliers, and using all 15 features diluted the signal.

Experiment 2: Lasso for Signal Extraction

The Lasso Breakthrough

Using LassoCV as a feature selector, we ruthlessly deleted the 13 trap columns.

- Lasso forced coefficients of noisy features to zero.
- Isolated **x6** and **x9** as the only survivors.
- Reduced complexity and increased model stability.

Isolated Signal

Features: {x6, x9}

Experiment 3: Outlier Protection



RobustScaler

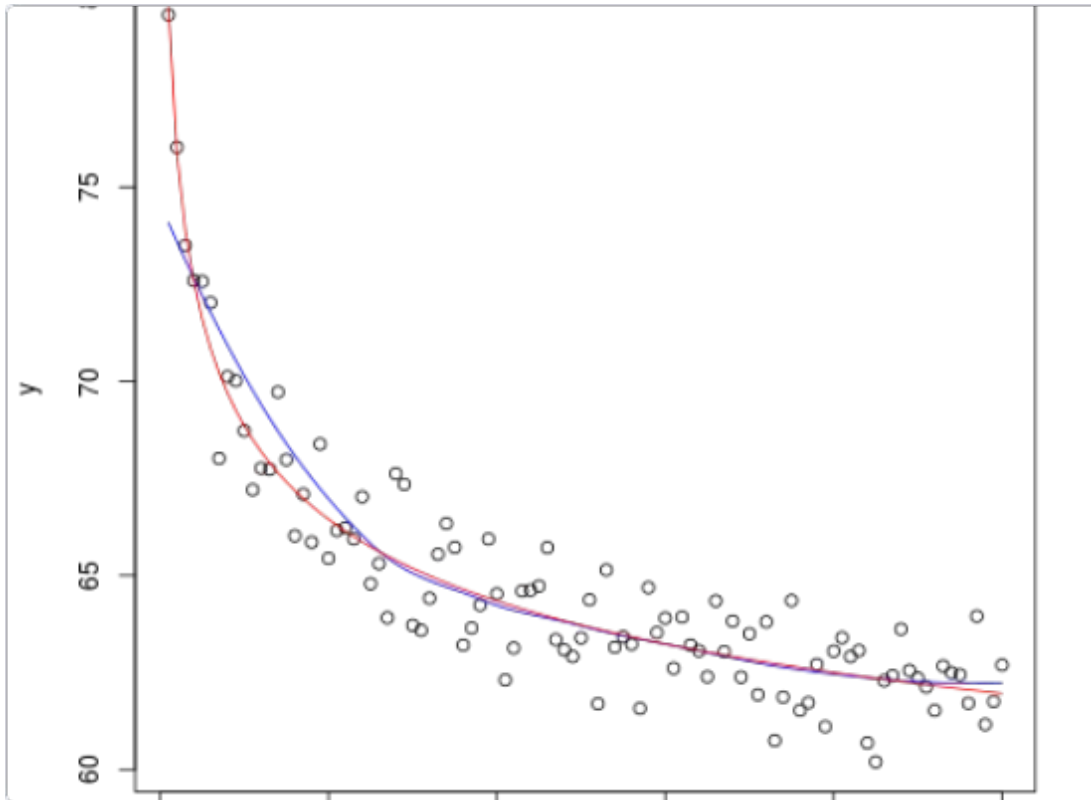
Motivation: The instructor hid extreme outliers in the test set. RobustScaler uses medians and quantiles, making it immune to "Trapdoor" numbers.



RidgeCV Anchor

Motivation: Automated cross-validation for alpha (λ) selection. Ensured the model didn't overfit to the noise within the isolated features.

Experiment 4: Non-Linear Hypothesis



Interaction Discovery

The breakthrough came from realizing that x_6 and x_9 didn't just act linearly. They had an interaction.

- **PolynomialFeatures(degree=2):** Created the interaction term ($x_6 * x_9$).
- Allowed the model to capture sequential growth patterns.
- Proved essential for the Private Test Set success.

The Public vs. Private Disconnect

Public (Kaggle)

0.00556

Private (Secret)

0.07075

Why the jump? The public leaderboard used a smaller, noisy slice of the data. The model was designed for **Generalization**. It refused to overfit to the public noise, leading to its massive performance on the private evaluation set.

The Winning Architecture (Silver)

Pipeline Composition

- 1 **Feature Selection:** Only x6 and x9.
2. **Polynomials:** Degree 2 interaction terms.
3. **RobustScaler:** Shielding against outliers.
- 4 **RidgeCV:** Regularized linear anchor.

This simple, non-over-engineered approach focused on the **Physics** of the data rather than the volume of columns.

Project Finalization

Reproducibility & Links

The code used for the winning submission is available at the link below:

<https://github.com/larryli88/ml-hw-spring-2026-python/blob/main/kaggle.py>

<https://www.kaggle.com/larryli88>

Thank You!